

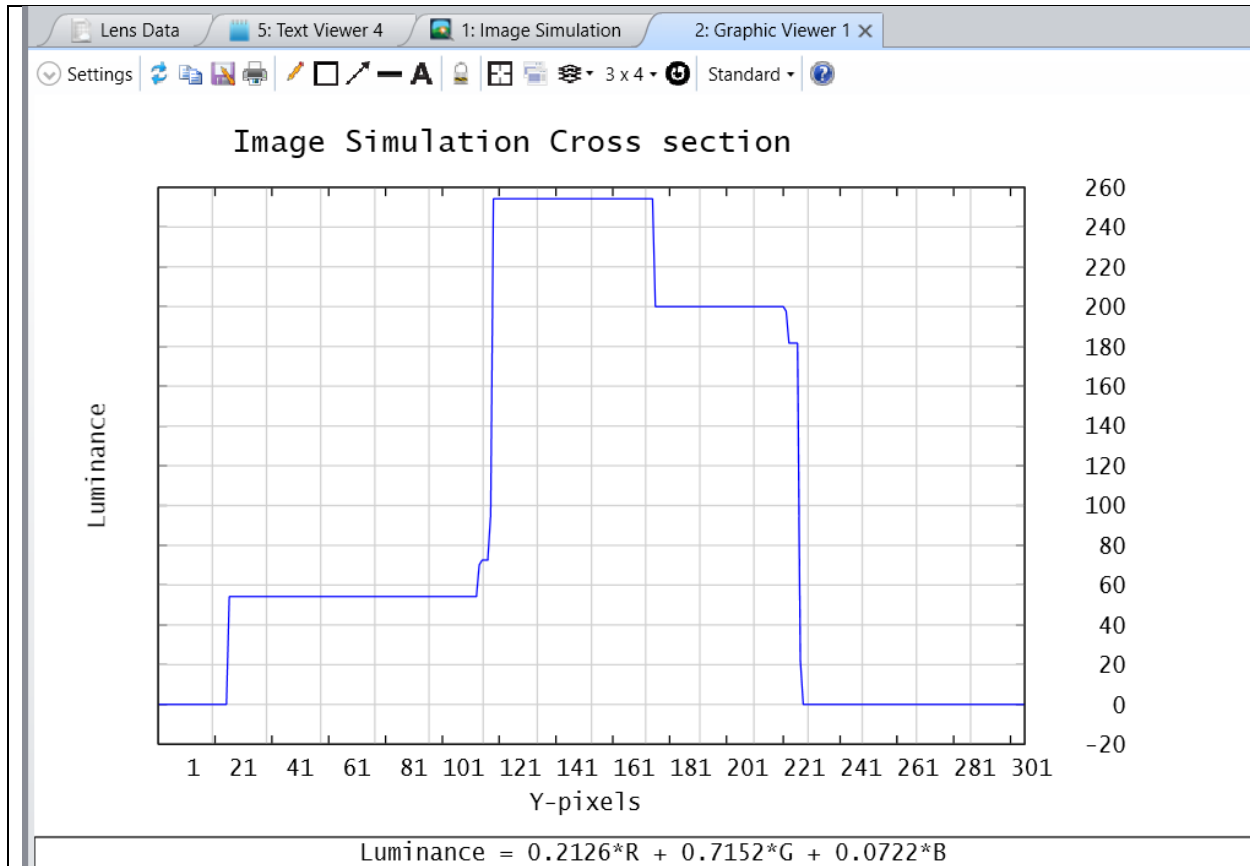
Title

ZPL_Macro_Sequential_Image_Simulation_Cross_Section

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Brief Description



Overview

This macro reads the text output of the image simulation. The text output contains the RGB values of the image for each pixel. It then creates a cross section of the image simulation in X or Y for a defined X or Y pixel. In the Image Simulation window, click on Save first before running it.

Plugin Type

ZPL Analysis

Language

ZPL

Updates

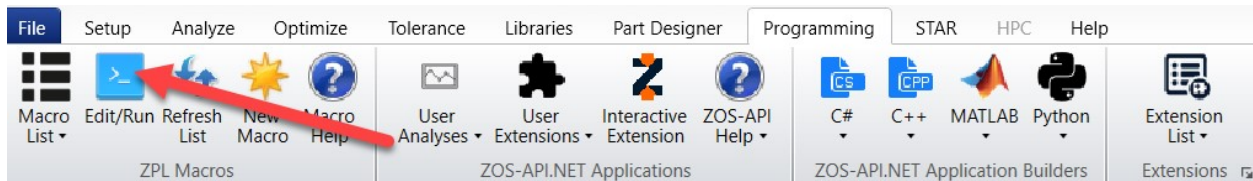
Date	Version	OpticStudio version	Comment
2021-12-09	1.0	-	Creation

Instructions

1. Installation

To run the macro, copy the .zpl file under {Zemax}\Macros.

The macro can then be run from:



2. How to use

The macro will run the image simulation and reads the RGB values.

It will then display a cross section: 4 plots are displayed.

- One for the R values
- One for the G values
- One for the B values
- One for the luminance values calculated from the RGB values.

The Luminance formula used here is:

Luminance = $0.2126 * R + 0.7152 * G + 0.0722 * B$

It comes from the following sources:

- [https://en.wikipedia.org/wiki/Luma_\(video\)](https://en.wikipedia.org/wiki/Luma_(video))
- <https://stackoverflow.com/questions/596216/formula-to-determine-perceived-brightness-of-rgb-color>
- <https://www.itu.int/rec/R-REC-BT.709>